1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI}:

$$\|\boldsymbol{A}\boldsymbol{r}_1(E_1) - \boldsymbol{B}\boldsymbol{r}_2(E_2)\| = r_{\text{SOI}} \tag{1}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i: semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i\sqrt{1 - e_i^2}$)
- E_i: eccentric anomaly of orbit i
- $\boldsymbol{A} \in \mathbb{R}^{3\times2}$, $\boldsymbol{B} \in \mathbb{R}^{3\times2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI}: radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\boldsymbol{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \tag{2}$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2)^\top(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2) = r_{\text{SOI}}^2 \tag{3}$$

Expand the left-hand side:

$$\boldsymbol{r}_1^\top\boldsymbol{A}^\top\boldsymbol{A}\boldsymbol{r}_1 - 2\boldsymbol{r}_1^\top\boldsymbol{A}^\top\boldsymbol{B}\boldsymbol{r}_2 + \boldsymbol{r}_2^\top\boldsymbol{B}^\top\boldsymbol{B}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{4}$$

1.4.1. Orthonormality of $\boldsymbol{A}$ and $\boldsymbol{B}$

Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}_{2\times2}, \quad \boldsymbol{B}^\top\boldsymbol{B} = \boldsymbol{I}_{2\times2} \tag{5}$$

Therefore the self-terms simplify:

$$\boldsymbol{r}_1^\top\boldsymbol{A}^\top\boldsymbol{A}\boldsymbol{r}_1 = \boldsymbol{r}_1^\top\boldsymbol{r}_1 = |\boldsymbol{r}_1|^2 =: r_1^2 \tag{6}$$

$$\boldsymbol{r}_2^\top\boldsymbol{B}^\top\boldsymbol{B}\boldsymbol{r}_2 = \boldsymbol{r}_2^\top\boldsymbol{r}_2 = |\boldsymbol{r}_2|^2 =: r_2^2 \tag{7}$$

1.4.2. The coupling matrix $\boldsymbol{C}$

Define the 2×2 matrix:

$$\boldsymbol{C} := \boldsymbol{A}^\top\boldsymbol{B} \tag{8}$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - 2\boldsymbol{r}_1^\top\boldsymbol{C}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{9}$$

with components:

$$\boldsymbol{C} = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} \boldsymbol{a}_1 \cdot \boldsymbol{b}_1 & \boldsymbol{a}_1 \cdot \boldsymbol{b}_2 \\ \boldsymbol{a}_2 \cdot \boldsymbol{b}_1 & \boldsymbol{a}_2 \cdot \boldsymbol{b}_2 \end{pmatrix} \tag{10}$$

where $\boldsymbol{a}_k$, $\boldsymbol{b}_k$ denote column k of $\boldsymbol{A}$, $\boldsymbol{B}$ respectively.

Note: $\boldsymbol{C}$ is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 1$.

1.5. Expanding the Scalar Terms

1.5.1. Heliocentric distances r_i^2

$$r_i^2 = a_i^2 (1 - e_i \cos E_i)^2 \tag{11}$$

1.5.2. Cross-term $\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2$

Write out the matrix product:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}^\top \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \tag{12}$$

where $x_i \coloneqq a_i (\cos E_i - e_i)$ and $y_i \coloneqq b_i \sin E_i$.

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = C_{11} x_1 x_2 + C_{12} x_1 y_2 + C_{21} y_1 x_2 + C_{22} y_1 y_2 \tag{13}$$

Substitute the explicit forms:

$$\begin{aligned} \boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = & C_{11} a_1 (\cos E_1 - e_1) \cdot a_2 (\cos E_2 - e_2) \\ & + C_{12} a_1 (\cos E_1 - e_1) \cdot b_2 \sin E_2 \\ & + C_{21} b_1 \sin E_1 \cdot a_2 (\cos E_2 - e_2) \\ & + C_{22} b_1 \sin E_1 \cdot b_2 \sin E_2 \end{aligned} \tag{14}$$

Expand each product:

$$(\cos E_1 - e_1)(\cos E_2 - e_2) = \cos E_1 \cos E_2 - e_2 \cos E_1 - e_1 \cos E_2 + e_1 e_2 \tag{15}$$

$$(\cos E_1 - e_1) \sin E_2 = \cos E_1 \sin E_2 - e_1 \sin E_2 \tag{16}$$

$$\sin E_1 (\cos E_2 - e_2) = \sin E_1 \cos E_2 - e_2 \sin E_1 \tag{17}$$

Collecting by type of dependence on E_1, E_2:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \underbrace{S(E_1, E_2)}_{\text{bilinear}} - \underbrace{e_1 Q(E_2)}_{E_2 \text{ only}} - \underbrace{e_2 P(E_1)}_{E_1 \text{ only}} + \underbrace{a_1 a_2 e_1 e_2 C_{11}}_{\text{constant}} \tag{18}$$

where:

$$S(E_1, E_2) \coloneqq C_{11} a_1 a_2 \cos E_1 \cos E_2 + C_{12} a_1 b_2 \cos E_1 \sin E_2 + C_{21} b_1 a_2 \sin E_1 \cos E_2 + C_{22} b_1 b_2 \sin E_1 \sin E_2 \tag{19}$$

$$P(E_1) \coloneqq C_{11} a_1 a_2 \cos E_1 + C_{21} b_1 a_2 \sin E_1 \tag{20}$$

$$Q(E_2) \coloneqq C_{11} a_1 a_2 \cos E_2 + C_{12} a_1 b_2 \sin E_2 \tag{21}$$

1.5.3. Full constraint

Substituting into Equation 9:

$$a_1^2 (1 - e_1 \cos E_1)^2 + a_2^2 (1 - e_2 \cos E_2)^2 - 2[S - e_1 Q - e_2 P + a_1 a_2 e_1 e_2 C_{11}] = r_{\mathrm{SOI}}^2 \tag{22}$$

Expand the squared terms and rearrange:

$$
\begin{aligned}
&\underbrace{\left(a_1^2 + a_2^2 - 2a_1 a_2 e_1 e_2 C_{11} - r_{\mathrm{SOI}}^2\right)}_{=:K} \\
&+\underbrace{\left(-2a_1^2 e_1 \cos E_1 + a_1^2 e_1^2 \cos^2 E_1 + 2e_2 P(E_1)\right)}_{=:f_1(E_1)} \\
&+\underbrace{\left(-2a_2^2 e_2 \cos E_2 + a_2^2 e_2^2 \cos^2 E_2 + 2e_1 Q(E_2)\right)}_{=:f_2(E_2)} \\
&-2S(E_1, E_2) = 0
\end{aligned}
\tag{23}
$$

1.5.4. Constant K

$$K := a_1^2 + a_2^2 - 2a_1 a_2 e_1 e_2 C_{11} - r_{\mathrm{SOI}}^2 \tag{24}$$

1.5.5. Terms depending only on E_1

Substituting Equation 20:

$$
\begin{aligned}
f_1(E_1) &= -2a_1^2 e_1 \cos E_1 + a_1^2 e_1^2 \cos^2 E_1 + 2e_2 (C_{11} a_1 a_2 \cos E_1 + C_{21} b_1 a_2 \sin E_1) \\
&= -2a_1 e_1 (a_1 - a_2 e_2 C_{11}) \cos E_1 + 2a_2 b_1 e_2 C_{21} \sin E_1 + a_1^2 e_1^2 \cos^2 E_1
\end{aligned}
\tag{25}
$$

1.5.6. Terms depending only on E_2

Substituting Equation 21:

$$f_2(E_2) = -2a_2 e_2 (a_2 - a_1 e_1 C_{11}) \cos E_2 + 2a_1 b_2 e_1 C_{12} \sin E_2 + a_2^2 e_2^2 \cos^2 E_2 \tag{26}$$

1.5.7. Compact form

$$K + f_1(E_1) + f_2(E_2) - 2S(E_1, E_2) = 0 \tag{27}$$

1.6. Half-Angle (Weierstrass) Substitution

To convert the transcendental equation into a polynomial, substitute:

$$t_i := \tan\left(\frac{E_i}{2}\right) \tag{28}$$

$$\cos E_i = \frac{1 - t_i^2}{1 + t_i^2}, \quad \sin E_i = \frac{2t_i}{1 + t_i^2} \tag{29}$$

This is valid for $E_i \neq \pi + 2\pi k$ (i.e. $t_i \neq \infty$), which corresponds to apoapsis. That case can be checked separately.

1.6.1. Rationalize f_1

Let $T_i := 1 + t_i^2$. From Equation 25, substitute Equation 29:

$$f_1 = -\lambda_1 \frac{1 - t_1^2}{T_1} + \sigma_1 \frac{t_1}{T_1} + q_1 \frac{\left(1 - t_1^2\right)^2}{T_1^2} \tag{30}$$

where we defined shorthand constants:

$$\lambda_1 := 2a_1 e_1 (a_1 - a_2 e_2 C_{11}), \quad \sigma_1 := 4a_2 b_1 e_2 C_{21}, \quad q_1 := a_1^2 e_1^2 \tag{31}$$

Multiply by T_1^2:

$$
\begin{aligned}
f_1 T_1^2 &= -\lambda_1 (1 - t_1^2)(1 + t_1^2) + \sigma_1 t_1 (1 + t_1^2) + q_1 \left(1 - t_1^2\right)^2 \\
&= -\lambda_1 (1 - t_1^4) + \sigma_1 (t_1 + t_1^3) + q_1 (1 - 2t_1^2 + t_1^4)
\end{aligned}
\tag{32}
$$

Collecting by powers of t_1:

$$f_1 T_1^2 = (q_1 - \lambda_1) + \sigma_1 t_1 - 2q_1 t_1^2 + \sigma_1 t_1^3 + (q_1 + \lambda_1) t_1^4 \tag{33}$$

1.6.2. Rationalize f_2

By analogy, define:

$$\lambda_2 := 2a_2 e_2 (a_2 - a_1 e_1 C_{11}), \quad \sigma_2 := 4a_1 b_2 e_1 C_{12}, \quad q_2 := a_2^2 e_2^2 \tag{34}$$

$$f_2 T_2^2 = (q_2 - \lambda_2) + \sigma_2 t_2 - 2q_2 t_2^2 + \sigma_2 t_2^3 + (q_2 + \lambda_2) t_2^4 \tag{35}$$

1.6.3. Rationalize S

From Equation 19, substituting Equation 29 for both variables:

$$\begin{aligned} S = C_{11} a_1 a_2 (1 - t_1^2) \frac{1 - t_2^2}{T_1 T_2} + C_{12} a_1 b_2 (1 - t_1^2) \frac{2t_2}{T_1 T_2} \\ + C_{21} b_1 a_2 (2t_1) \frac{1 - t_2^2}{T_1 T_2} + C_{22} b_1 b_2 (2t_1) \frac{2t_2}{T_1 T_2} \end{aligned} \tag{36}$$

$$\begin{aligned} S \cdot T_1 T_2 = C_{11} a_1 a_2 &(1 - t_1^2 - t_2^2 + t_1^2 t_2^2) \\ &+ 2C_{12} a_1 b_2 (t_2 - t_1^2 t_2) \\ &+ 2C_{21} b_1 a_2 (t_1 - t_1 t_2^2) \\ &+ 4C_{22} b_1 b_2 t_1 t_2 \end{aligned} \tag{37}$$

1.6.4. Full polynomial constraint (distance)

Multiply Equation 27 by $T_1^2 T_2^2$:

$$K T_1^2 T_2^2 + (f_1 T_1^2) T_2^2 + T_1^2 (f_2 T_2^2) - 2(S \cdot T_1 T_2) T_1 T_2 = 0 \tag{38}$$

Each factor in parentheses has been computed in Equation 33, Equation 35, Equation 37. This is a bivariate polynomial of bidegree $(4, 4)$ in (t_1, t_2):

$$F(t_1, t_2) = 0 \tag{39}$$

1.7. Structure of the Quartic in t_1

For a fixed t_2, Equation 38 is a quartic in t_1:

$$F(t_1; t_2) = c_4 t_1^4 + c_3 t_1^3 + c_2 t_1^2 + c_1 t_1 + c_0 = 0 \tag{40}$$

where each c_j is a polynomial of degree ≤ 4 in t_2.

1.7.1. Explicit coefficients

To collect the coefficients, first extract the t_1-structure of each contribution to Equation 38.

***S* contribution.** From Equation 37, collect by powers of t_1:

$$S \cdot T_1 T_2 = s_0 + s_1 t_1 - s_0 t_1^2 \tag{41}$$

where:

$$\begin{aligned} s_0 &:= C_{11} a_1 a_2 (1 - t_2^2) + 2C_{12} a_1 b_2 t_2 \\ s_1 &:= 2C_{21} b_1 a_2 (1 - t_2^2) + 4C_{22} b_1 b_2 t_2 \end{aligned} \tag{42}$$

Multiply by $(1 + t_1^2)$ to form $-2(S \cdot T_1 T_2) T_1 T_2$:

$$(s_0 + s_1 t_1 - s_0 t_1^2)(1 + t_1^2) = s_0 + s_1 t_1 + s_1 t_1^3 - s_0 t_1^4 \tag{43}$$

Remaining terms. $KT_1^2T_2^2$ and $T_1^2(f_2T_2^2)$ are even in t_1 via the factor $T_1^2 = 1 + 2t_1^2 + t_1^4$, contributing at powers t_1^0, t_1^2, t_1^4 with ratio 1 : 2 : 1. Define:

$$\Gamma \coloneqq f_2T_2^2 = (q_2 - \lambda_2) + \sigma_2 t_2 - 2q_2t_2^2 + \sigma_2 t_2^3 + (q_2 + \lambda_2)t_2^4 \tag{44}$$

Summing all four contributions by powers of t_1:

$$c_0 = (KT_2^2 + q_1 - \lambda_1)T_2^2 + \Gamma - 2s_0T_2 \tag{45}$$

$$c_1 = \sigma_1 T_2^2 - 2s_1T_2 \tag{46}$$

$$c_2 = (2KT_2^2 - 2q_1)T_2^2 + 2\Gamma \tag{47}$$

$$c_4 = (KT_2^2 + q_1 + \lambda_1)T_2^2 + \Gamma + 2s_0T_2 \tag{48}$$

One can verify that $c_0 + c_4 - c_2 = 4q_1T_2^2$, confirming Equation 51.

1.7.2. The $c_1 = c_3$ identity

The quartic **always** satisfies $c_1 = c_3$, regardless of orbital parameters. This follows from the structure of each contribution to Equation 38:

- $KT_1^2T_2^2$ and $T_1^2(f_2T_2^2)$: contain only **even** powers of t_1 (since $T_1^2 = (1 + t_1^2)^2$ and f_2 is independent of t_1). They contribute nothing to c_1 or c_3.
- $f_1T_1^2$: from Equation 33, the coefficients of t_1 and t_1^3 are both σ_1, so $c_1 = c_3$ for this part.
- $S \cdot T_1T_2$: is at most degree 2 in t_1 (from Equation 37). When multiplied by $T_1T_2 = (1 + t_1^2)(1 + t_2^2)$, the factor $(1 + t_1^2)$ guarantees equal t_1^1 and t_1^3 coefficients: if $S \cdot T_1T_2 = s_0 + s_1t_1 + s_2t_1^2$, then $(s_0 + s_1t_1 + s_2t_1^2)(1 + t_1^2) = s_0 + s_1t_1 + (s_0 + s_2)t_1^2 + s_1t_1^3 + s_2t_1^4$.

Therefore the quartic has the form:

$$c_4t_1^4 + c_1t_1^3 + c_2t_1^2 + c_1t_1 + c_0 = 0 \tag{49}$$

1.7.3. Decomposition

This can be split into a biquadratic part (even powers only) and a residual:

$$F(t_1; t_2) = \underbrace{c_4t_1^4 + c_2t_1^2 + c_0}_{=:B(t_1^2)} + c_1t_1(t_1^2 + 1) \tag{50}$$

The biquadratic $B(w) = c_4w^2 + c_2w + c_0$ is a quadratic in $w = t_1^2$.

Polynomial division by $(t_1^2 + 1)$ gives:

$$F = (t_1^2 + 1)(c_4t_1^2 + c_1t_1 + c_2 - c_4) + \underbrace{(c_0 + c_4 - c_2)}_{=4q_1T_2^2} \tag{51}$$

where $q_1 = a_1^2e_1^2$ from Equation 31.

1.7.4. Special cases

The remainder in Equation 51 vanishes when $q_1 = 0$, i.e. when $e_1 = 0$ (circular spacecraft orbit). In that case $(t_1^2 + 1)$ is a factor; its roots $t_1 = \pm i$ are complex, so the quartic reduces to the **quadratic** $c_4t_1^2 + c_1t_1 + c_2 - c_4 = 0$, giving at most 2 real geometric solutions per E_2.

When the orbital planes are coplanar with identical orientation ($\boldsymbol{C} = \boldsymbol{I}$), we have $C_{21} = 0$ and $C_{12} = 0$, which makes $\sigma_1 = 0$ (from Equation 31). Since $c_1 = \sigma_1T_2^2 + \dots$ terms proportional to C_{21} and C_{22} cross-products, coplanarity drives $c_1 \to 0$. With $c_1 = 0$, Equation 50 reduces to the pure biquadratic $B(t_1^2) = 0$, solvable via a single quadratic formula in t_1^2.

Condition	Simplification	Effective degree
General	$c_1 = c_3$ quartic	4
Circular s/c ($e_1 = 0$)	$(t_1^2 + 1)$ factors out $\to$ quadratic	2
Coplanar ($\boldsymbol{C} = \boldsymbol{I}$)	$c_1 = 0 \to$ biquadratic	2 (in t_1^2)
Both	Trivial quadratic	2

In the general (non-degenerate) case, $c_1 = c_3$ does not reduce the quartic to lower degree, and Ferrari's method (or equivalent) is required.

1.7.5. Ferrari's resolvent cubic

To solve the quartic Equation 49 via Ferrari's method, first convert to depressed form by substituting $t_1 = u - c_1/(4c_4)$:

$$u^4 + \alpha u^2 + \beta u + \gamma = 0 \tag{52}$$

where:

$$\begin{aligned} \alpha &:= \frac{c_2}{c_4} - 3\frac{c_1^2}{8c_4^2} \\ \beta &:= \frac{c_1}{c_4}\left(1 - \frac{c_2}{2c_4} + \frac{c_1^2}{8c_4^2}\right) \\ \gamma &:= \frac{c_0}{c_4} - \frac{c_1^2}{4c_4^2} + c_1^2\frac{c_2}{16c_4^3} - 3\frac{c_1^4}{256c_4^4} \end{aligned} \tag{53}$$

The $c_1 = c_3$ identity manifests as the factored form of β: the linear coefficient of the depressed quartic is proportional to c_1. When $c_1 = 0$ (coplanar case), $\beta = 0$ and the depressed quartic is a biquadratic — consistent with the special-case analysis above.

Ferrari's method rewrites Equation 52 as:

$$\left(u^2 + \frac{y}{2}\right)^2 = (y - \alpha)u^2 - \beta u + \left(\frac{y^2}{4} - \gamma\right) \tag{54}$$

The right-hand side is a perfect square in u when its discriminant vanishes, giving the resolvent cubic:

$$y^3 - \alpha y^2 - 4\gamma y + \left(4\alpha\gamma - \beta^2\right) = 0 \tag{55}$$

The constant term $4\alpha\gamma - \beta^2$ vanishes when $\beta = 0$ (i.e. $c_1 = 0$), making $y = 0$ a root of the resolvent — from which Ferrari immediately yields the biquadratic factorization, consistent with the coplanar special case.

Once a real root y_0 of Equation 55 is found (analytically via Cardano's formula or numerically), define:

$$m := \sqrt{y_0 - \alpha}, \quad n := \frac{\beta}{2m} \tag{56}$$

(the case $m = 0$ corresponds to $\beta = 0$, i.e. the biquadratic reduction). The depressed quartic factors into two quadratics:

$$\left(u^2 + mu + \frac{y_0}{2} - n\right)\left(u^2 - mu + \frac{y_0}{2} + n\right) = 0 \tag{57}$$

Each factor is a standard quadratic, giving up to 4 roots for u, from which $t_1 = u - c_1/(4c_4)$.

For numerical implementation, the resolvent cubic should be solved by a robust cubic solver (handling the casus irreducibilis when all three roots are real). The subsequent quadratic solves are straightforward.

1.8. Time-Equality Constraint

Both bodies must be at their respective positions **at the same time**. This is the second equation of the system.

1.8.1. Kepler's equation

For orbit i, the time t relates to eccentric anomaly via Kepler's equation:

$$n_i(t - \tau_i) = E_i - e_i \sin E_i \tag{58}$$

where $n_i = \sqrt{\mu / a_i^3}$ is the mean motion, τ_i is the epoch of perihelion passage, and μ is the gravitational parameter of the central body.

1.8.2. Eliminating time

From Equation 58, express the common time t from each orbit and equate:

$$\tau_1 + \frac{E_1 - e_1 \sin E_1}{n_1} = \tau_2 + \frac{E_2 - e_2 \sin E_2}{n_2} \tag{59}$$

Rearrange:

$$\frac{E_1 - e_1 \sin E_1}{n_1} - \frac{E_2 - e_2 \sin E_2}{n_2} = \tau_2 - \tau_1 \tag{60}$$

Since E_i is an angle (defined modulo 2π), the general form accounting for multiple orbital revolutions (k_1, k_2 complete orbits) is:

$$\frac{E_1 - e_1 \sin E_1 + 2\pi k_1}{n_1} - \frac{E_2 - e_2 \sin E_2 + 2\pi k_2}{n_2} = \tau_2 - \tau_1 \tag{61}$$

Define the epoch offset constant:

$$\Delta_\tau \coloneqq \tau_2 - \tau_1 \tag{62}$$

Multiply through by $n_1 n_2$:

$$n_2(E_1 - e_1 \sin E_1 + 2\pi k_1) - n_1(E_2 - e_2 \sin E_2 + 2\pi k_2) = n_1 n_2 \Delta_\tau \tag{63}$$

1.8.3. Separation into analytical and transcendental parts

Rearrange Equation 63:

$$n_2 E_1 - n_1 E_2 - n_2 e_1 \sin E_1 + n_1 e_2 \sin E_2 = n_1 n_2 \Delta_\tau - 2\pi(n_2 k_1 - n_1 k_2) \tag{64}$$

The right-hand side is a constant for each choice of integer pair (k_1, k_2). Define:

$$\Phi(k_1, k_2) \coloneqq n_1 n_2 \Delta_\tau - 2\pi(n_2 k_1 - n_1 k_2) \tag{65}$$

So the time-equality constraint is:

$$G(E_1, E_2) \coloneqq n_2 E_1 - n_1 E_2 - n_2 e_1 \sin E_1 + n_1 e_2 \sin E_2 - \Phi = 0 \tag{66}$$

Note: G is **not** rationalizable via the Weierstrass substitution because it contains both E_i and $\sin E_i$ — the E_i terms are transcendental in $t_i = \tan(E_i/2)$. This is the fundamental reason Kepler's equation has no closed-form inverse.

1.9. The Full System

The SOI encounter problem is the 2×2 nonlinear system:

$$\begin{cases} F(E_1, E_2) = 0 & \text{(distance constraint, polynomial in } t_1 = \tan(E_1/2),\ t_2 = \tan(E_2/2)) \\ G(E_1, E_2) = 0 & \text{(time equality, transcendental)} \end{cases} \tag{67}$$

for unknowns $(E_1, E_2) \in [0, 2\pi) \times [0, 2\pi)$, parametrized by the integer pair (k_1, k_2).

1.10. Restricting the Search Domain

Before solving the full system, we can drastically narrow the range of E_1 (and symmetrically E_2) that needs to be searched.

1.10.1. Coarse filter: heliocentric radius bounds

For a fixed spacecraft position at E_1, the heliocentric distance is $r_1 = a_1(1 - e_1 \cos E_1)$. The planet's heliocentric distance ranges over $r_2 \in [a_2(1 - e_2), a_2(1 + e_2)]$ as E_2 varies. By the triangle inequality, the minimum possible distance between the two bodies (over all E_2 and all mutual orientations) satisfies:

$$d_{\text{min}} \geq |r_1 - r_2| \tag{68}$$

A necessary condition for the spacecraft to be within r_{SOI} of the planet is therefore $|r_1 - r_2| \leq r_{\text{SOI}}$ for **some** r_2 in the planet's range. This requires:

$$a_2(1 - e_2) - r_{\text{SOI}} \leq r_1 \leq a_2(1 + e_2) + r_{\text{SOI}} \tag{69}$$

Substituting $r_1 = a_1(1 - e_1 \cos E_1)$ and solving for $\cos E_1$:

$$\frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \leq \cos E_1 \leq \frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \tag{70}$$

(with the inequality direction flipped when $e_1 < 0$, but $e_1 \geq 0$ by definition). Clamp both sides to $[-1, 1]$ before taking arccos. If the interval is empty, no encounter is possible on this orbit pair. If it covers $[-1, 1]$ entirely, no pruning occurs (only possible when the orbits nearly overlap).

For typical interplanetary trajectories, $r_{\text{SOI}} \ll a_2$, and the spacecraft orbit crosses the planet's radial range over a narrow arc. This filter eliminates the vast majority of E_1 values. For example, Voyager 2's heliocentric orbit crosses Jupiter's radial range (approximately 4.95–5.46 AU) over a small fraction of its eccentric anomaly.

1.10.2. Fine filter: quartic real-root existence

Within the interval surviving Equation 70, a tighter test asks: does the quartic $F(t_1, t_2) = 0$ (viewed as a quartic in t_2 for fixed $t_1 = \tan(E_1/2)$) have at least one real root?

For a given E_1, the quartic in t_2 has the same $c_1 = c_3$ structure (by the symmetry of the argument in the section on the quartic structure, applied with the roles of t_1 and t_2 exchanged). Its coefficients are pure numbers once t_1 is fixed. The quartic has real roots if and only if there exists a $t_2 \in \mathbb{R}$ (corresponding to $E_2 \in [0, 2\pi) \setminus .\{\pi\}$) satisfying it.

Rather than evaluating the full quartic discriminant (which is algebraically unwieldy), a practical approach is:

- **Evaluate F at a few sample t_2 values.** If F changes sign, a real root exists by the intermediate value theorem.
- **Check the minimum of F over t_2.** Since F is a degree-4 polynomial in t_2 with positive leading coefficient (for non-degenerate orbits), it has a global minimum. If $F_{\text{min}} \leq 0$, real roots exist. The minimum can be found by solving $\partial F / \partial t_2 = 0$ — a cubic in t_2, solvable analytically.

This identifies the exact set of E_1 values for which the SOI sphere (centered at some planet position) geometrically intersects the spacecraft orbit. The boundary of this set — where the quartic transitions from having real roots to having none — corresponds to the quartic having a double root (tangency of the SOI sphere with the orbit ellipse).

1.11. Solution Procedure

1.11.1. Step 1: Enumerate orbit-count pairs

Choose a time window $[t_{\min}, t_{\max}]$. For each orbit, this bounds the number of complete revolutions:

$$k_i \in \left\{ \left\lfloor \frac{t_{\min} - \tau_i}{T_i} \right\rfloor, \ldots, \left\lceil \frac{t_{\max} - \tau_i}{T_i} \right\rceil \right\} \tag{71}$$

where $T_i = 2\pi / n_i$ is the orbital period. For each pair (k_1, k_2), we solve the system Equation 67 with the corresponding $\Phi(k_1, k_2)$ from Equation 65.

1.11.2. Step 2: Restrict the E_1 domain

Apply the coarse heliocentric radius filter Equation 70 to obtain a candidate interval (or intervals) $\mathcal{I}_1 \subset [0, 2\pi)$. Optionally refine with the quartic real-root existence test to tighten $\mathcal{I}_1$ further.

1.11.3. Step 3: Solve the distance quartic

For each sampled $E_1 \in \mathcal{I}_1$, compute $t_1 = \tan(E_1/2)$ and solve the quartic Equation 49 in t_2 (or equivalently, for each sampled E_2, solve the quartic in t_1). Use Ferrari's method, or the quadratic/biquadratic reductions from the special cases above when applicable, yielding up to 4 real branches $t_2^{(j)}(t_1)$, i.e. up to 4 curves $E_2(E_1)$ satisfying the distance constraint.

1.11.4. Step 4: Substitute into the time constraint

Along each branch $E_2^{(j)}(E_1)$, substitute into Equation 66 to get a single-variable equation:

$$g^{(j)}(E_1) \coloneqq G\left(E_1, E_2^{(j)}(E_1)\right) = 0 \tag{72}$$

This reduces the 2×2 system to a scalar root-finding problem. The function $g^{(j)}$ is continuous but transcendental (due to the E_i and $\sin E_i$ terms from Kepler's equation).

1.11.5. Step 5: Root finding on each branch

Sweep E_1 over $\mathcal{I}_1$ on each branch j and locate sign changes of $g^{(j)}(E_1)$. At each sign change, apply a scalar root-finding method (e.g. Brent's method or the ITP method) to find E_1^* such that $g^{(j)}(E_1^*) = 0$. For each root E_1^*, recover $E_2^* = 2 \arctan\left(t_2^{(j)}(\tan(E_1^*/2))\right)$.

1.11.6. Step 6: Newton refinement

Ferrari's quartic formula is numerically unstable: it involves nested square and cube roots and suffers from catastrophic cancellation, particularly when roots are nearly degenerate or when polynomial coefficients span many orders of magnitude (typical in orbital mechanics, where distances in SI units produce coefficients differing by > 20 orders). In practice, Ferrari often yields only $\sim$ 8–10 correct digits out of 16. Near branch bifurcation points (where roots merge or split during the E_1 sweep), the quartic has near-double roots and accuracy degrades further.

To recover full precision, apply Newton's method on the original 2×2 system Equation 67 using the candidate (E_1^*, E_2^*) from Step 5 as initial guess. The Jacobian is:

$$\boldsymbol{J} = \begin{pmatrix} \frac{\partial F}{\partial E_1} & \frac{\partial F}{\partial E_2} \\ \frac{\partial G}{\partial E_1} & \frac{\partial G}{\partial E_2} \end{pmatrix} \tag{73}$$

From Equation 66:

$$\frac{\partial G}{\partial E_1} = n_2(1 - e_1 \cos E_1), \quad \frac{\partial G}{\partial E_2} = -n_1(1 - e_2 \cos E_2) \tag{74}$$

These are proportional to r_i / a_i, which is strictly positive for elliptical orbits ($e_i < 1$).

From Equation 27:

$$\frac{\partial F}{\partial E_1} = f_1'(E_1) - 2\frac{\partial S}{\partial E_1}, \quad \frac{\partial F}{\partial E_2} = f_2'(E_2) - 2\frac{\partial S}{\partial E_2} \tag{75}$$

where, from Equation 25:

$$f_1' = 2a_1 e_1 (a_1 - a_2 e_2 C_{11}) \sin E_1 + 2a_2 b_1 e_2 C_{21} \cos E_1 - a_1^2 e_1^2 \sin 2E_1 \tag{76}$$

and from Equation 19:

$$\frac{\partial S}{\partial E_1} = -C_{11} a_1 a_2 \sin E_1 \cos E_2 - C_{12} a_1 b_2 \sin E_1 \sin E_2 + C_{21} b_1 a_2 \cos E_1 \cos E_2 + C_{22} b_1 b_2 \cos E_1 \sin E_2 \tag{77}$$

(and analogously for $\partial F / \partial E_2$).

One or two Newton iterations typically suffice to reach machine precision.

1.11.7. Step 7: Recover encounter time

For each refined solution (E_1^*, E_2^*), the encounter time is:

$$t^* = \tau_i + \frac{E_i^* - e_i \sin E_i^* + 2\pi k_i}{n_i} \tag{78}$$

(using either $i = 1$ or $i = 2$ — both give the same t^* at a true solution).

1.12. Summary

1. **Precompute**: $\boldsymbol{C} = \boldsymbol{A}^\top \boldsymbol{B}$, constants K, λ_i, σ_i, q_i, mean motions n_i, epoch offset Δ_τ.
2. **Enumerate**: Loop over integer orbit pairs (k_1, k_2) within the desired time window.
3. **Restrict domain**: Apply heliocentric radius bounds Equation 70 (and optionally the quartic real-root existence test) to obtain a narrow candidate interval $\mathcal{J}_1 \subset [0, 2\pi)$.
4. **Distance polynomial**: For each (k_1, k_2), construct the bidegree-$(4, 4)$ polynomial Equation 38 in (t_1, t_2).
5. **Branch extraction**: For each sampled $E_1 \in \mathcal{J}_1$, solve the quartic in t_2 analytically, obtaining up to 4 branches $E_2(E_1)$.
6. **Scalar root-finding**: On each branch, find zeros of $g^{(j)}(E_1) = G\left(E_1, E_2^{(j)}(E_1)\right) = 0$ via sign-change detection and Brent's/ITP method.
7. **Newton refinement**: Polish each candidate on the full 2×2 system Equation 67 to recover precision lost by Ferrari's formula.
8. **Recover time**: Compute encounter epoch via Equation 78.